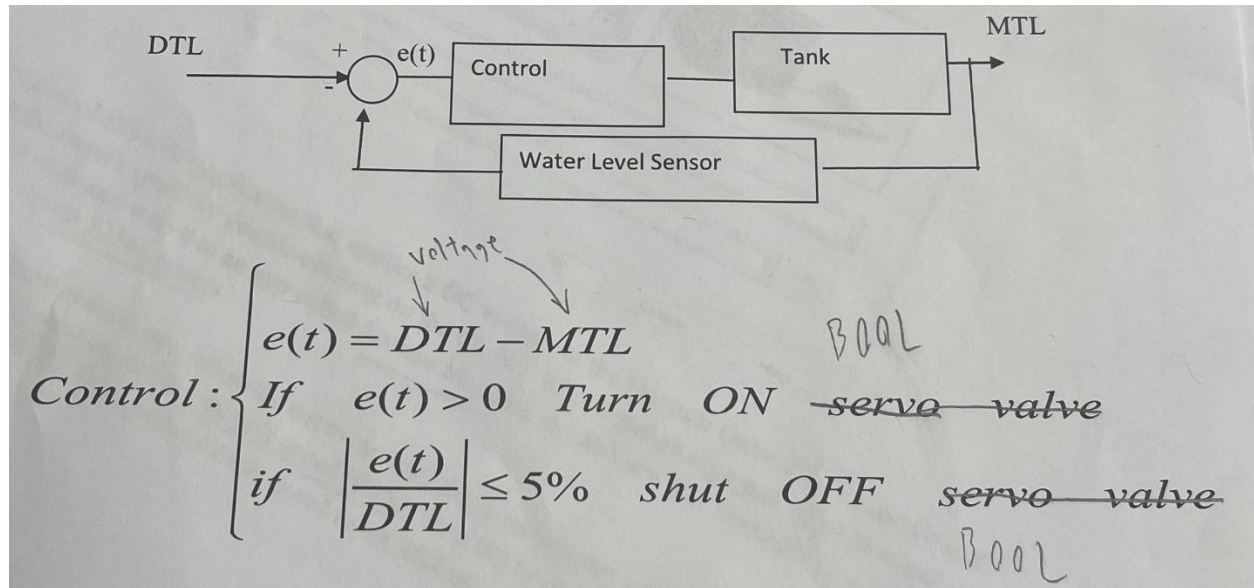


## Introduction

This lab was mislabeled as Experiment 6 on the given paper, but it was actually Experiment 8.

The first task completed was Part B, and it was a water control system. According to the TAs we did not have a servo valve to control, so we controlled a Boolean variable instead following the logic below:



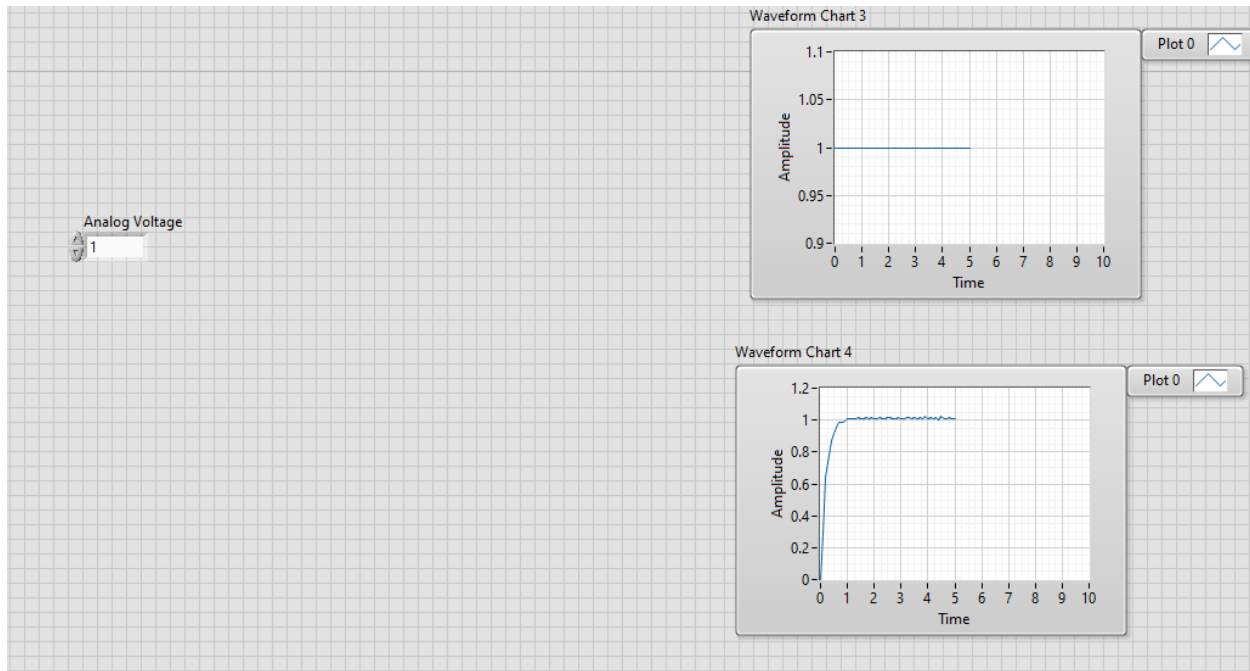
Using the BOOL as an LED in the front panel we would verify if we got our logic correct.

We also did a redo of Lab8a because we incorrectly modelled the motor. The motor is controlled using an analog voltage input and increases in speed as this voltage increases. I primarily followed the diagram to us in Lab to model my Block Diagram. HIL initialize and CL HIL Read is used to read the output speed of the motor, and CL HIL Write are used to input a voltage to the motor.

# Analysis

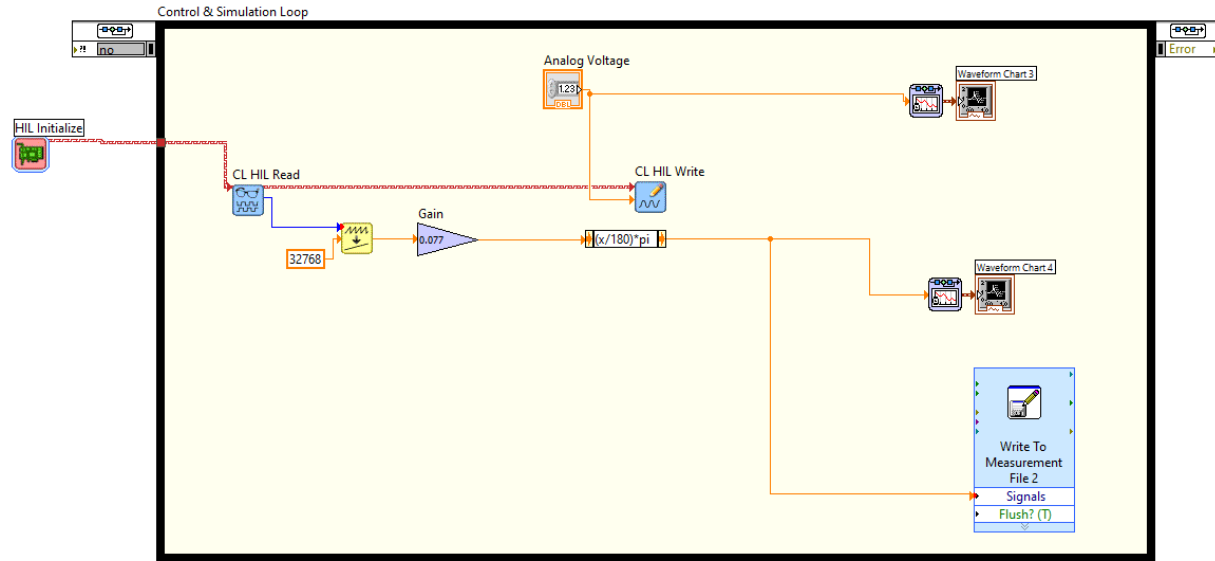
## Part A – Motor Control

Front Panel:



This SimTime charts are made in the block diagram and enable me to see the time as a parameter rather than some random data from a waveform chart. Notably the analog voltage stays constant while the output voltage reaches a steady state approximately equal to the input voltage.

## Block Diagram:



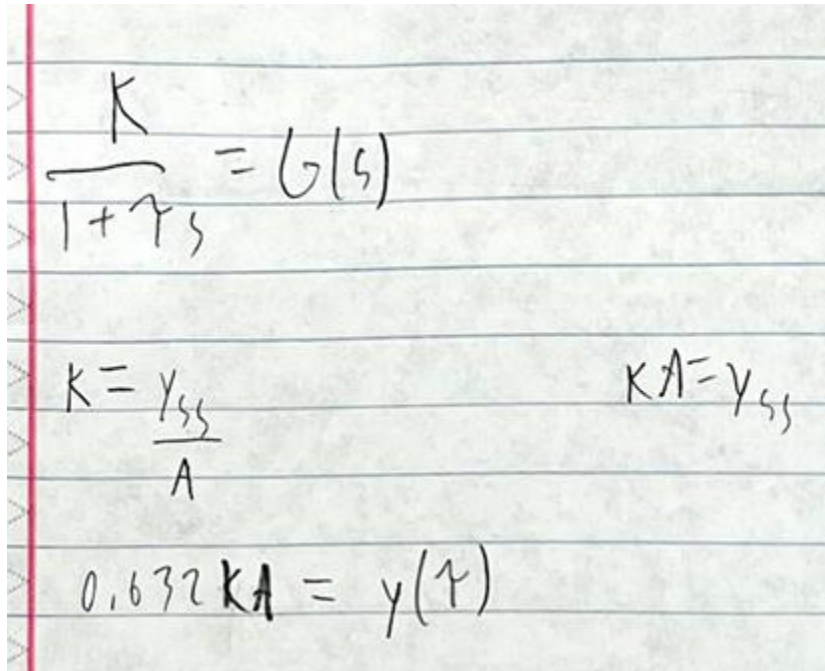
Ignore the Write to Measurement file, it is unnecessary. I took the data by hand instead.

The inverse modulus used with the constant 32768 is to convert rpm to speed. The gain is the gain of the motor, and the  $(x/180) \cdot \pi$  is to convert from degrees to radians. The sensor reads this speed as an output voltage.

Data Collected:

I conducted separate experiments for 1 V stepping up to 10 V in 1 V increments.

I knew I could model the transfer function using the following form:



The image shows three handwritten equations on lined paper. The first equation is  $\frac{K}{1 + \tau s} = G(s)$ . The second equation is  $K = \frac{y_{ss}}{A}$ . The third equation is  $KA = y_{ss}$ . Below these, the equation  $0.632 KA = y(\tau)$  is written.

$$\frac{K}{1 + \tau s} = G(s)$$
$$K = \frac{y_{ss}}{A}$$
$$KA = y_{ss}$$
$$0.632 KA = y(\tau)$$

Where  $K$  is a constant,  $\tau$  is the time constant, and  $A$  is the constant multiplied by the unit step function,  $y_{ss}$  is the steady state output.

In this experiment,  $A$  is the Analog Input Voltage and it was changed by the operator using a numeric indicator.

$K$  was calculated by looking at the graph's steady state output and dividing by the associated  $A$ .

To find  $\tau$ , I multiplied  $y_{ss}$  (also known as  $KA$ ) by 0.632. Then I looked for that value on the y-axis and looked for the associated time value on the x-axis. That gave me my  $\tau$ .

I obtained the following results:

| Analog Voltage | $y_{ss}$ | $0.632 y_{ss}$ | $\tau$ | $G(s)$                  |
|----------------|----------|----------------|--------|-------------------------|
| 1              | 1        | 0.632          | 0.2    | $\frac{1}{1+0.2s}$      |
| 2              | 2        | 1.264          | 0.2    | $\frac{1}{1+0.2s}$      |
| 3              | 3.1      | 1.959          | 0.2    | $\frac{1.03}{1+0.2s}$   |
| 4              | 4.1      | 2.596          | 0.2    | $\frac{1.025}{1+0.2s}$  |
| 5              | 5.2      | 3.286          | 0.2    | $\frac{1.04}{1+0.2s}$   |
| 6              | 6.2      | 3.914          | 0.2    | $\frac{1.03}{1+0.2s}$   |
| 7              | 7.2      | 4.55           | 0.2    | $\frac{1.03}{1+0.2s}$   |
| 8              | 8.2      | 5.18           | 0.2    | $\frac{1.025}{1+0.2s}$  |
| 9              | 9.6      | 6.0672         | 0.18   | $\frac{1.067}{1+0.18s}$ |
| 10             | 10.2     | 6.4464         | 0.2    | $\frac{1.02}{1+0.2s}$   |

By inspection, all the steady state outputs are almost identical to their associated  $A$ s, so:

$$K \approx 1$$

Likewise, almost every  $\tau$  was about 0.2 except for when  $A = 9$ .

$$\tau \approx 0.2$$

The transfer function for each of these is identical with some minor error. This is due to the measurement uncertainty of the sensors used to communicate with the servo motor.

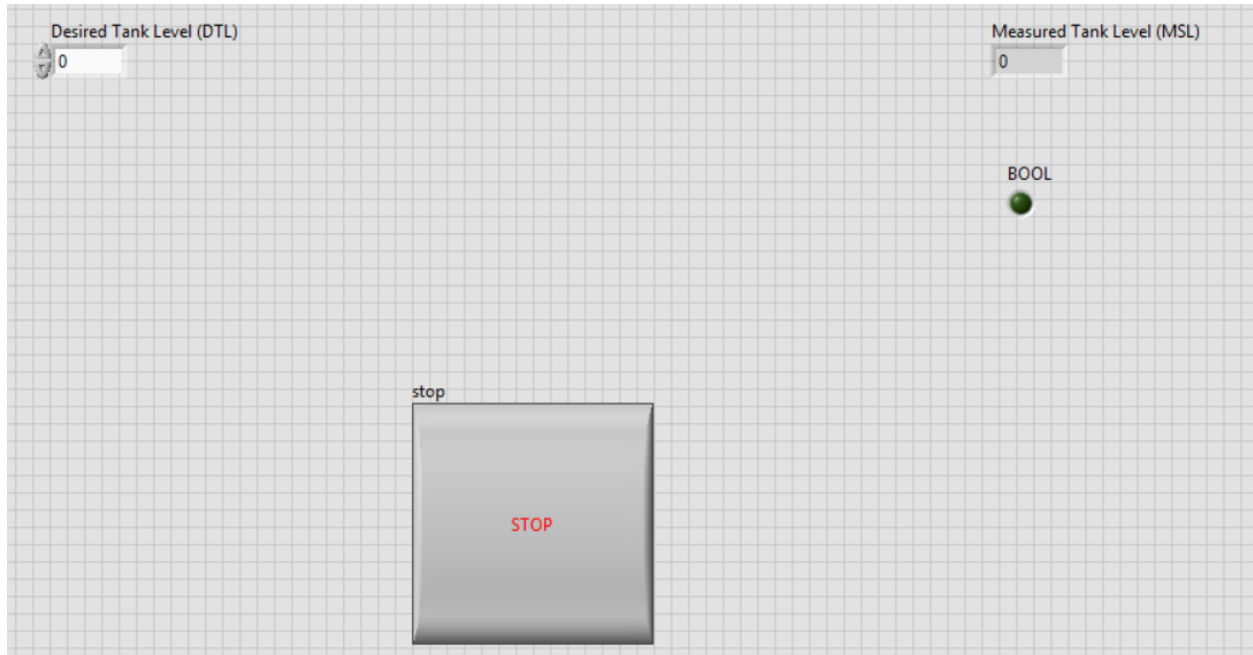
Approximating  $K$  and  $\tau$ , a practical transfer function of the motor is:

$$G(s) = \frac{1}{1+0.2s}$$

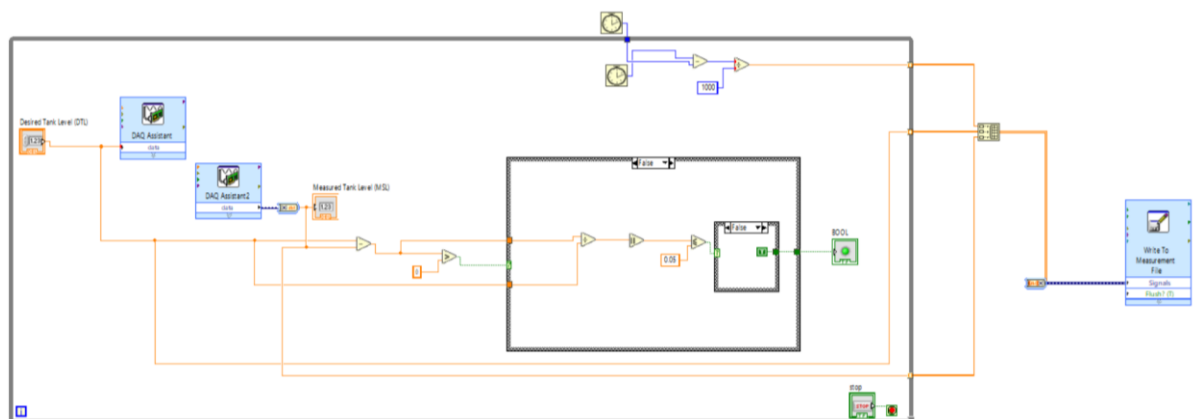
### Part B – Water Control:

As desired by the lab TAs, I used a BOOL instead of a servo valve. This meant the water tank would not be directly controlled by the servo valve, but the bool would be on whenever the servo valve should be and off when the servo valve should not.

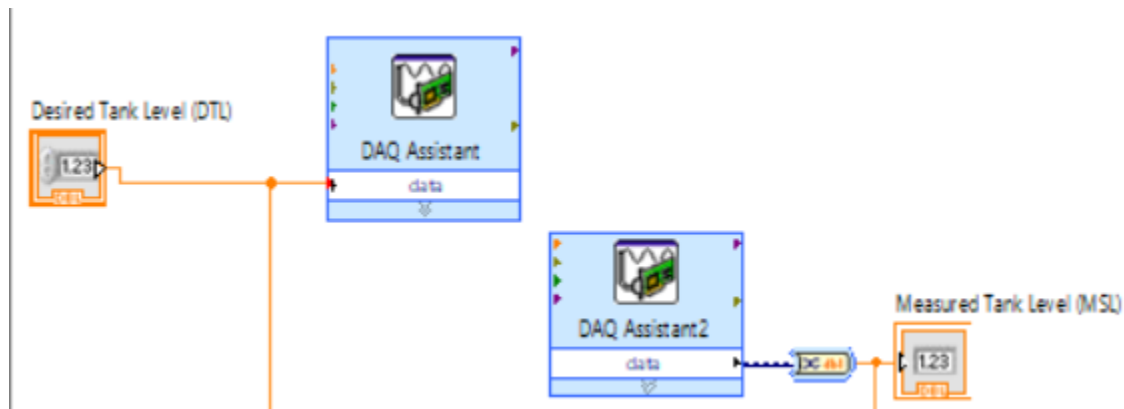
Front Panel:



Block Diagram:



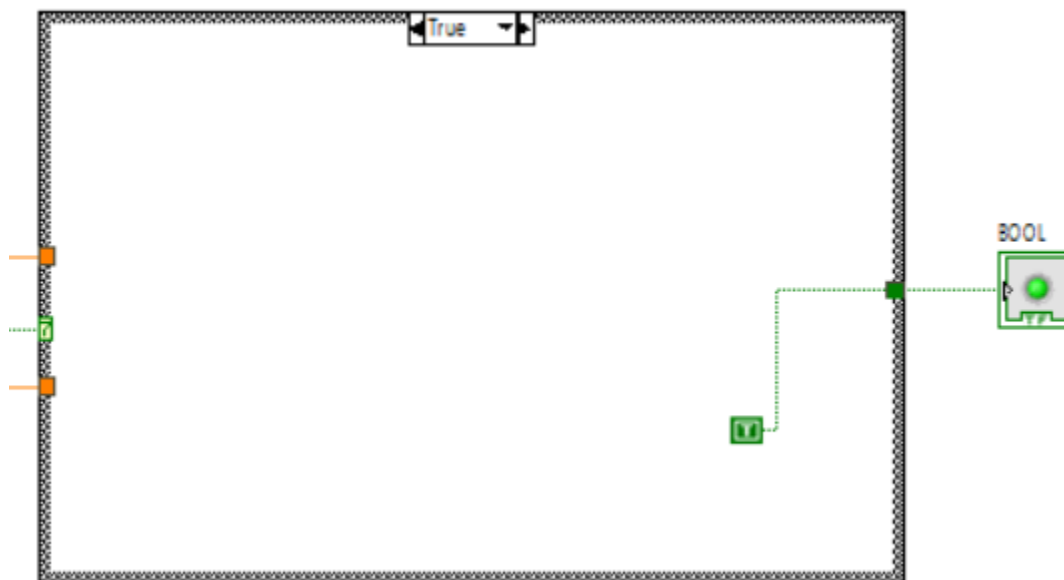
DAQ:



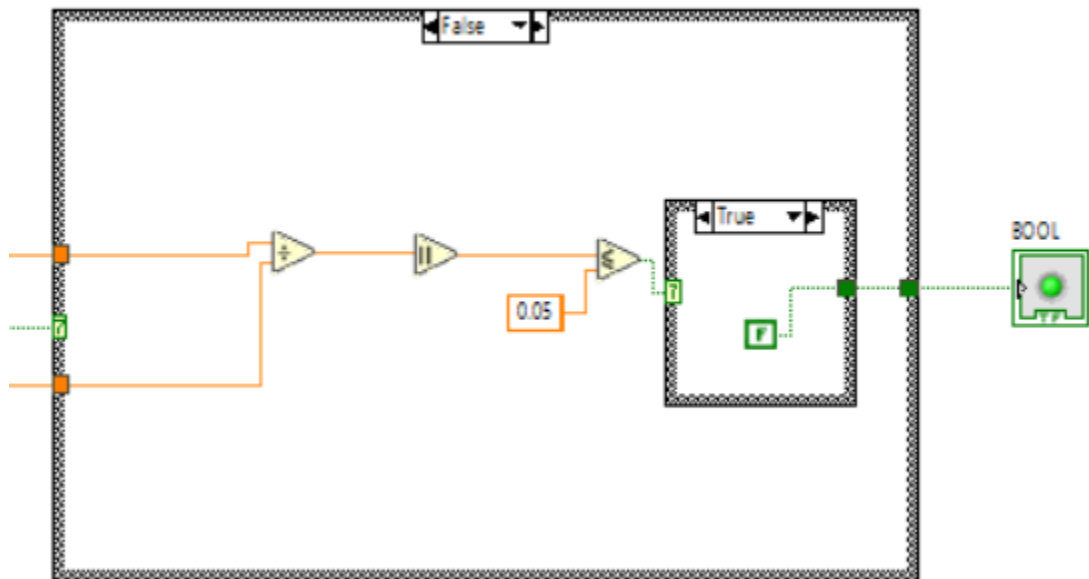
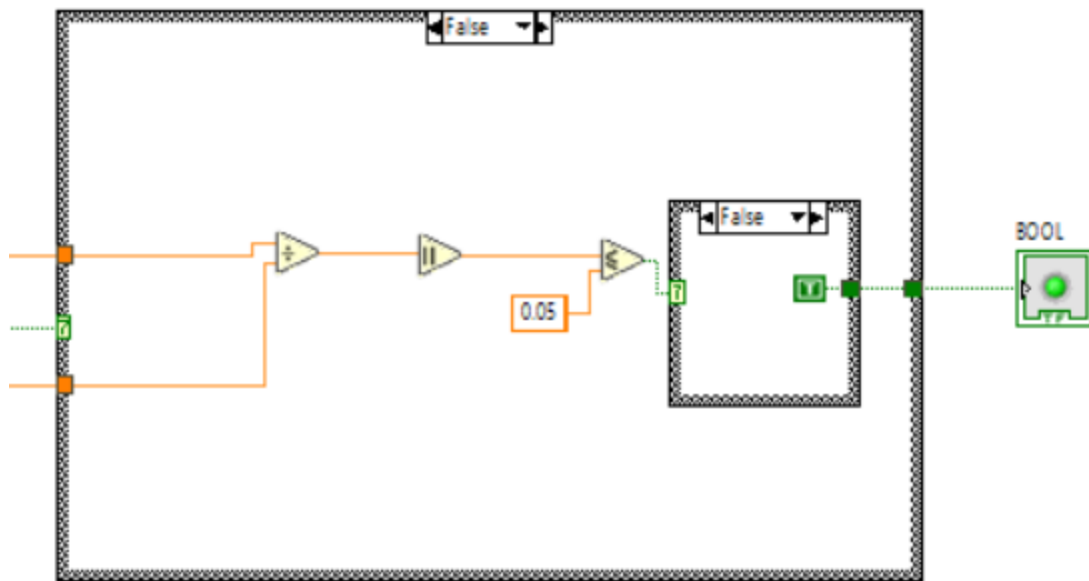
The DAQs remained mostly the same from the previous Lab 7. The main difference is that the output voltage is used to show the measured tank level. I used the data collected from the previous lab to know what the DTL voltage should be for certain tank levels (1.3 V for a level 10, which is what I used to test my BOOL).

Case Structure:

This is the overarching case that detects if  $e(t) > 0$  to turn on the BOOL.

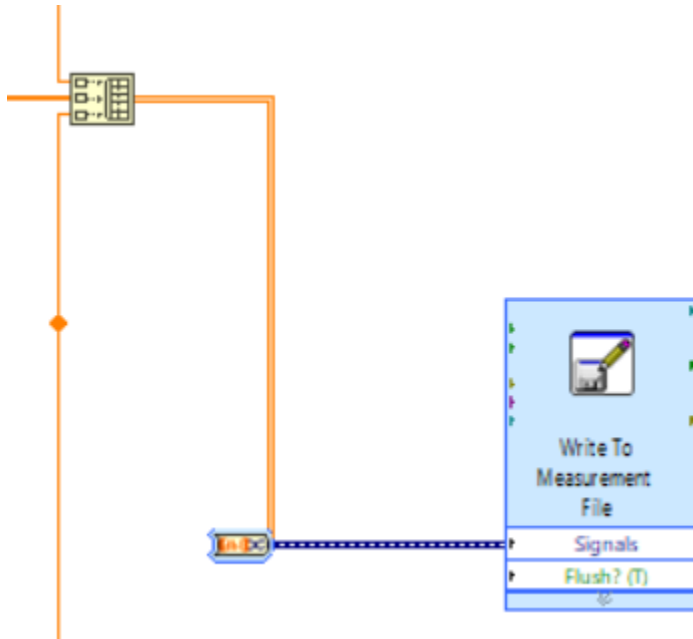


If this is not the case, it will check if for the other case of if the absolute value of  $e(t)$  divided by DTL is less than or equal to 5%



When this statement evaluates true, the BOOL is off. Otherwise, BOOL stays true.

Saving to File:



This is a simple implementation to build an array then place it in an Excel file, which keeps it a bit better organized than a delimited spreadsheet.

I collected the time, control signal (DTL voltage) and the water tank level (MTL voltage).

| Time  | Control Signal (DTL) | Water Tank Level (MTL) |
|-------|----------------------|------------------------|
| 0     | 1.3                  | 1.663732               |
| 0.021 | 1.3                  | 1.663404               |
| 0.043 | 1.3                  | 1.662417               |

Notably this data is not very useful without the visual aspect of the LED.

In an actual real-world situation where there was actual servo valve rather than a BOOL, this would be useful for Chemical processing plants where the level of the amounts of chemicals is very important. Regulating the desired amounts of chemicals in those containers can alter the chemical processes dramatically, so it is important to have a level control.

## Conclusion

Part A Redo was a good application of what we learned in lecture for finding transfer functions, primarily calculating  $K$  and  $\tau$ . This will be useful for the coming labs as we do more with the PIDs and unit step response. Part B was an interesting application to see how controlling a water tank may work.